1. Project Planning & Management

1.1 Project Overview and Objectives

The Retail Demand & Sales Intelligence Platform is an end-to-end Data Science and Business Intelligence solution designed to address key retail analytics and demand forecasting challenges. Retailers often have large volumes of historical sales data but may struggle to transform this data into actionable business insights, understand sales patterns, and make informed decisions about future demand.

The primary objective of this project is to develop an integrated analytics, forecasting, and prediction system that transforms historical retail data into meaningful insights and accessible predictions. The platform will combine the following components:

- **Data Preparation and SQL Database Management:** Clean and preprocess the dataset, store relevant data in a relational SQL database, and support multidimensional queries across dates, stores, product families, and regions where available.
- **Exploratory Data Analysis (EDA):** Analyze sales trends, product performance, seasonal patterns, and relevant relationships within the dataset.
- **Machine Learning and Sales Forecasting:** Develop and compare regression models to forecast sales, using appropriate evaluation metrics and a time-based validation strategy.
- **Interactive Business Intelligence Dashboard:** Visualize key performance indicators (KPIs), sales trends, product comparisons, and forecasting results to support data-driven business decisions.
- **Prediction REST API:** Develop a model-serving API using FastAPI or Flask to accept valid prediction inputs, such as supported store and

product-family features, and return sales forecasts from the trained model.

- **Business Insights and Recommendations:** Translate analytical findings and forecasting results into practical recommendations that may support demand planning, inventory management, and retail performance monitoring.

The overall goal is to integrate data management, analytics, machine learning, visualization, and API-based model access into a unified solution that demonstrates how retail data can support more informed business decisions.

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The project covers the development of an end-to-end retail sales analytics and forecasting prototype. Its scope includes data cleaning and preprocessing, relational database design using SQL, exploratory data analysis, machine learning model development and evaluation, an interactive dashboard for visualizing sales insights, and a prediction REST API for generating forecasts from supported input features.

The project focuses on historical data analysis and forecasting. Full production deployment, real-time integration with external retail systems, and automated inventory ordering are outside the current scope.

1.2 Dataset Selection

The project is planned around Kaggle's Store Sales – Time Series Forecasting dataset, which contains historical grocery sales information from Favorita stores in Ecuador. The dataset includes sales records and supporting information about stores, transactions, holidays, promotions, and oil prices. The team will confirm the final set of files and fields after an initial data inspection.

Dataset source: https://www.kaggle.com/competitions/store-sales-time-series-forecasting/data

Important data note: the competition's test.csv file does not contain target sales values. Model development and evaluation will therefore use a chronological split of the labeled historical data rather than treating the competition test file as a validation set.

1.3 Project Timeline and Milestones

The project timeline and documentation milestones are aligned with the official DEPI Industry Track schedule, as follows:

- Project Planning & Management Submission: 2 October 2026
- Literature Review Submission: 9 October 2026
- Requirements Gathering Submission: 9 October 2026
- System Analysis & Design Submission: 9 October 2026
- Implementation (Source Code & Execution): 16 October 2026
- Final Presentation, Testing & Reports: 16 October 2026

GitHub Repository Management: The repository will be maintained throughout the project using version control. It will contain the source code, organized notebooks, SQL database scripts, prediction API implementation, and project documentation.

1.4 Team Structure, Task Allocation, and Responsibilities

To ensure seamless execution and accountability, the project responsibilities are divided among six team members based on key technical modules:

Team Member	Detailed Responsibilities	Assigned Module
Belal Mostafa Atta	Handles data ingestion, missing value imputation, duplicate removal, outlier handling, and data type formatting to	Data Cleaning & preprocessing

	prepare a clean dataset for SQL storage and modeling.	
Ahmed Essam Abdelmonem	Designs the relational database schema, writes SQL scripts for database creation and data insertion, and creates reusable SQL queries/views to analyze KPIs by product, store, region, and date.	SQL Database Management
Raheel Romany Abdelmasih	Conducts exploratory data analysis, analyzes seasonal/temporal trends, evaluates high- and low-performing products, and extracts actionable business insights.	EDA & Business insights
Kirollos Safwat shawky	Designs and implements feature engineering pipelines, including lag features, rolling averages, calendar features, and other relevant predictors. Trains and tunes multiple regression models, such as Linear Regression, Random Forest, and Gradient Boosting or XGBoost where feasible. Collaborates with Member 5 to improve feature selection, model configurations, and training workflows.	Feature Engineering & Regression Modeling
Mariam Sayed Abdullah	Supports Member 4 during model development by researching suitable algorithms, suggesting relevant features, helping optimize model configurations, and reviewing training results. Independently evaluates and compares trained models using MAE and RMSE, with MAPE included when appropriate. Implements chronological validation strategies, analyzes prediction errors, checks for data leakage, compares model performance against a baseline, and recommends the best-performing model for final	Model Evaluation, Optimization & validation

	integration. Documents evaluation results and communicates improvement opportunities to Member 4.	
Elbasel Waleed Hassan	Dashboard Development, Prediction API, and System Integration Develops the interactive dashboard using Streamlit, Dash, or Power BI to present sales KPIs, trends, business insights, and forecasts. Implements a prediction REST API using FastAPI or Flask in collaboration with Members 4 and 5, enabling the trained model to generate forecasts from valid supported inputs. Integrates the dashboard, API, trained model, and data components; organizes the GitHub repository; and prepares integration tests and deployment where feasible within the project timeline.	Dashboard , API & System Integration

1.5 Risk Management and Mitigation Strategies

Mitigation Strategy	Impact / Likelihood	Identified Risk	Risk Category
Apply systematic data cleaning and validation, check missing values and duplicates, verify data types, and investigate outliers before preprocessing is finalized.	High / Medium	Missing, inconsistent, or invalid sales records in the dataset.	Data Quality
Define consistent data formats and API request/response schemas. Conduct regular integration tests	High / Low	Incompatibility between database outputs, ML models, and the	Technical Integration

and validate data flow between system components.		dashboard frontend.	
Use chronological validation or time-series cross-validation, tune hyperparameters carefully, compare models against a naive baseline, and prevent future-data leakage.	Medium / Medium	Regression models fitting noise in training data, resulting in poor test metrics.	Model Overfitting
Hold weekly team meetings, define internal deadlines and task dependencies, track progress using GitHub Projects, and address delays early.	High / Low	Delays in individual module handoffs impacting dependent tasks.	Timeline Delays
Validate incoming requests, use a consistent feature schema, handle invalid inputs with clear error messages, and test the API using valid and invalid sample requests.	Medium / Medium	Invalid prediction inputs or mismatches between the API input features and the trained model.	API Reliability

1.6 Key Performance Indicators (KPIs) for Project Success

To evaluate the success of the project implementation, the following measurable performance indicators are established:

- **Functional Completion:** Successful implementation and integration of all core project components, including data preprocessing, SQL database management, exploratory data analysis, machine learning, the interactive dashboard, and the prediction REST API.

- **Model Performance:** Evaluation of forecasting models using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The selected model should outperform a standard baseline model on the

same chronological validation dataset using consistent evaluation procedures.

- **Dashboard Performance:** Target dashboard query and visualization rendering time of under 3 seconds under normal testing conditions.

- **API Performance and Reliability:** Target response time of under 500 milliseconds for valid prediction requests under normal local testing conditions. The API should return valid predictions and handle invalid inputs with appropriate error messages.

- **Documentation and Code Quality:** A complete and organized GitHub repository containing clean source code, well-structured notebooks, SQL database scripts, model files or reproducible model-training instructions, API usage documentation, project setup instructions, and full project documentation.

- **Business Insight Delivery:** Identification and presentation of meaningful sales trends, product performance patterns, and demand forecasting insights, accompanied by practical recommendations that support retail planning and inventory management.

2. Literature Review

2.1 Related Research Studies

Retail demand forecasting and sales analytics help businesses understand historical performance, identify demand patterns, and make better decisions about inventory and sales planning. The impact of external factors on sales, machine learning forecasting techniques, and the usage of interactive dashboards to assist business information have all been studied in the past. Additionally, real-world projects have combined machine learning, business intelligence, SQL, and exploratory data analysis to create retail analytics systems.

Study 1: From Data to Decisions: Optimizing Supply Chain Management with Machine Learning-Infused Dashboards (2024)

Based on actual transaction data from an Austrian business-to-business retailer of hygiene products, Zimmermann and Brandtner offer an applied analytics solution. RFM analysis, market basket analysis, TURF analysis, and demand forecasting are among the machine learning and data analysis methods used in this study. Interactive dashboards that facilitate corporate intelligence and decision-making are used to display the outcomes.

Relevance to the Proposed Project: This study shows how interactive dashboards and predictive analytics can be combined to provide a workable business solution. In the same way, the proposed project integrates dashboard visualization, demand forecasting, and sales analysis, with a focus on SQL queries and systematic regression model comparison.

Strengths: The study uses real business data and integrates several analytical methods with interactive visualizations. This makes the solution relevant to practical business decision-making.

Limitations: Its scope includes broader supply chain and customer analytics. It does not specifically establish the same workflow of relational database querying, systematic regression model comparison, and sales analysis by the dimensions required in the proposed project.

Source: https://doi.org/10.1016/j.procs.2024.05.184

Study 2: Comparative Study of AutoML and Machine Learning in Retail Forecasting (2025)

Kristian and Suri use over 27,300 transaction records from a traditional Indonesian business to compare automated and traditional machine learning techniques. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) are used in the study to evaluate models and AutoML frameworks. The findings show that while several AutoML frameworks provide different levels of accuracy and automation, SARIMAX does very well on the dataset.

Relevance to the Proposed Project: This study supports the importance of evaluating forecasting approaches using consistent performance metrics rather than selecting a model without testing its accuracy.

Strengths: It compares multiple forecasting approaches using recognized evaluation metrics and discusses how seasonality and data characteristics affect forecasting performance.

Limitations: The study's primary focus is on automation and predicting performance. Developing a comprehensive analytics platform that combines relational SQL analysis, exploratory analysis, interactive dashboards, and business reporting is not its main goal.

Source: https://doi.org/10.1016/j.procs.2025.09.031

Study 3: Applying Machine Learning in Retail Demand Prediction: A Comparison of Tree-Based Ensembles and Long Short-Term Memory-Based Deep Learning (2023)

For retail demand prediction, Nasseri, Falatouri, Brandtner, and Darbanian compared LSTM-based deep learning with tree-based ensemble regressors. In addition to using external variables like COVID-19 and meteorological data, the study includes more than six years of daily demand data for more than 330 products, totaling 5.2 million records from a well-known retail business. The tree-based ensembles are compared with LSTM-based models, and Extra Trees Regressors are evaluated in combination with XGBoost, Random Forest Regression, and Gradient Boosting Regression.

Relevance to the Proposed Project: This study directly supports comparing several regression approaches on retail demand data, which is a core component of the proposed project.

Strengths: It uses a large, real-world, product-level dataset and compares multiple regression ensembles against a deep learning alternative.

Limitations: Its focus is model comparison and forecasting rather than a complete analytics platform combining relational SQL analysis, dashboards, and business reporting. Its dataset size and deep learning component may also exceed the scope of a small academic project.

Source: https://doi.org/10.3390/app131911112

2.2 Similar Practical Projects and Solutions

Project 1: Retail Intelligence & Demand Forecasting — TAMANGP2

An end-to-end retail analytics and demand forecasting system using Python, PostgreSQL, SQL, machine learning, and Power BI is shown in this GitHub project. It creates reusable SQL analytics views, processes 536,641 retail

transactions, and evaluates product and customer performance. Its forecasting pipeline assesses a Random Forest model, four-week and eight-week moving averages, and a Naive model. It creates four-week demand estimates for twenty products and chooses the strategy with the lowest MAE for each product. A three-page interactive Power BI dashboard is used to display the results.

Relevance to the Proposed Project: This is a close practical example because it integrates a relational database, SQL analytics, forecasting evaluation, and Power BI in a single workflow.

Strengths: The project demonstrates an end-to-end data pipeline, reusable SQL views, automated selection of the lowest-MAE forecasting approach, and interactive dashboards. Forecast results are also stored in PostgreSQL.

Limitations: Random Forest, a machine learning regression model, is the only one of its evaluated methods; the others are straightforward forecasting baselines. Its forecasting pipeline employs MAE as its documented selection metric and focuses on 20 product categories. Additional models, confidence ranges, and planned data refreshes are identified by the repository as potential enhancements.

Project link: https://github.com/TAMANGP2/retail-intelligence-demand-forecasting

Project 2: Retail Demand Forecasting & Inventory Planning — adityakarle

This project uses historical data from February 2010 to October 2012 to predict weekly retail sales across 45 Walmart locations. It verifies the validity of the data using SQLite, looking for null values, invalid negative sales numbers, and duplicate store-date combinations. Additionally, it trains a Random Forest regressor, generates calendar and lag-based features, and performs exploratory analysis. A time-based test set is used to compare the model with a naive forecasting baseline. An interactive Power BI dashboard is created by exporting forecast results.

Relevance to the Proposed Project: This project closely matches the demand forecasting workflow and demonstrates how data validation, EDA, model evaluation, and Power BI can be combined.

Strengths: It uses a time-based train/test split, evaluates the model against a baseline, and provides store-level visual analysis. The reported MAPE is 4.46% for Random Forest, compared with 4.71% for the naive baseline.

Limitations: Instead of evaluating many regression techniques, the project assesses a single regression model against a naive baseline. Rather than acting as the primary relational database for sales analytics, SQLite is utilized for data-quality validation. A live inventory management system is not the solution because the dataset does not include actual inventory or stock-level data.

Project link: https://github.com/adityakarle/retail-demand-forecasting

2.3 Research Gap and Project Motivation

Retail forecasting, SQL-based analytics, machine learning, and interactive dashboards are proven methods for business decision support, as the evaluated research and real-world applications show. While real-world projects combine database operations, forecasting, and visualization, some research concentrate on predicting accuracy or the impact of extra characteristics.

However, the practical projects reviewed provide a more specific opportunity for differentiation. The TAMANGP2 project evaluates Naive and moving-average baselines alongside one regression model, Random Forest. The adityakarle project also compares one regression model with a naive baseline.

Among the practical projects reviewed, forecasting evaluation centers on a single regression model (Random Forest) compared against naive or moving-average baselines. Our platform instead compares several

regression approaches using multiple evaluation metrics and supports querying sales by product, category, store, region, and date.

2.4 Expected Contribution

The proposed Retail Demand & Sales Intelligence Platform is expected to provide the following contributions:

1. A structured relational database for retail sales data.
2. SQL queries for business KPIs and analysis by product family, store, region, and date.
3. Exploratory analysis of sales trends, seasonal patterns, and business performance.
4. A comparison of selected regression models using multiple forecasting metrics.
5. An interactive dashboard presenting historical sales and forecast results.
6. A business report containing data-driven findings and recommendations.
7. An optional API extension, depending on project feasibility and timeline.

These contributions are intended to provide a practical and reproducible workflow that connects historical sales data with demand forecasts and business decision support.

3. Requirements Gathering

This section defines the stakeholders, user interactions, functional capabilities, and quality expectations for the Retail Demand & Sales Intelligence Platform.

3.1 Stakeholder Analysis

Stakeholder analysis identifies the people who will use, manage, or benefit from the proposed system and defines their main needs.

Stakeholder	Role	Needs and Expectations
Retail Management	**Makes business decisions**	**Understand revenue, sales performance, product performance, and future demand.**
Sales Analysts	**Analyze historical sales data**	**Filter data, calculate KPIs, perform EDA, and identify sales trends.**
Store Managers	**Monitor individual store performance**	**Compare sales across stores and regions and identify high- and low-performing products.**
Data Scientist	**Develop and evaluate predictive models**	**Prepare data, build regression models, compare model performance, and generate predictions.**

System Administrator / Developer	**Maintains the platform**	**Manage the database, maintain system functionality, and resolve technical issues.**

3.2 User Stories and Use Cases

A. User Stories

ID	User Story	Priority
US-01	As a manager, I want to view total revenue and sales KPIs so that I can evaluate overall business performance.	High
US-02	As a sales analyst, I want to filter sales by product, category, store, region, and date so that I can investigate specific business areas.	High
US-03	As a store manager, I want to identify the best- and worst-performing products so that I can make better product decisions.	High
US-04	As a sales analyst, I want to explore historical sales patterns so that I can	High

	identify seasonal and temporal trends.	
US-05	As a data scientist, I want to train and compare multiple regression models so that I can select a suitable model for predicting a numerical sales-related target.	High
US-06	As a manager, I want to view an interactive dashboard so that I can understand business performance through charts and visualizations.	High
US-07	As a developer, I want to store sales data in a relational database so that data can be organized, queried, and analyzed consistently.	High
US-08	As a manager, I want to read a business report summarizing findings and recommendations so that I can make data-driven decisions.	Medium

B. Use Case Descriptions

1: Analyze Sales Performance

Primary actor: Sales Analyst / Manager **Priority:** High

Preconditions

- Historical sales data is available.
- The database and analytical components are accessible.

Main flow

- The user opens the dashboard.
- The system retrieves the relevant sales data.
- The user selects filters such as date, product, category, store, or region.
- The system updates the displayed KPIs and charts.
- The user examines the results to identify sales trends and performance differences.

Expected outcome: The user obtains a clear view of sales performance for the selected criteria.

2: Generate Sales Predictions

Primary actor: Data Scientist **Priority:** High

Preconditions

- Historical data has been prepared.
- A numerical prediction target has been defined.

Main flow

- The data scientist prepares the relevant dataset.
- The data is divided into training and testing sets using an appropriate strategy.

- Multiple regression approaches are trained.
- The models are evaluated using suitable metrics.
- Their results are compared, and the most suitable model is selected.

Expected outcome: A trained regression model and documented evaluation results are available for the defined prediction problem.

3: Identify Product and Seasonal Trends

Primary actor: Sales Analyst / Manager **Priority:** High

Main flow

- The user selects a time period or business category.
- The system displays relevant sales summaries and visualizations.
- The user compares product and regional performance.
- The user examines changes across days, months, or other available time intervals.

Expected outcome: High- and low-performing products and potential seasonal patterns are identified.

3.3 Functional Requirements

Functional requirements specify what the system must do. The requirements below are derived from the project scope.

ID	Requirement	Priority
FR-01	The system shall store historical retail sales data in a relational database.	Must
FR-02	The system shall retrieve and query sales data by	Must

	product, category, store, region, and date.	
FR-03	The system shall calculate total revenue and relevant sales KPIs.	Must
FR-04	The system shall support Exploratory Data Analysis (EDA) to understand data distributions, relationships, and quality.	Must
FR-05	The system shall identify high- and low-performing products based on defined sales performance measures.	Must
FR-06	The system shall analyze historical sales patterns over available time periods to investigate seasonality and temporal trends.	Must
FR-07	The project shall support the development of a regression model for a defined numerical prediction target.	Must
FR-08	The system shall compare multiple regression approaches using	Must

	appropriate evaluation metrics.	
FR-09	The system shall provide an interactive dashboard containing relevant KPIs, filters, charts, and visualizations.	Must
FR-10	The project shall produce a business report summarizing analytical findings and actionable recommendations.	Must
FR-11	The project shall maintain the source code, notebook, and supporting documentation in a GitHub repository.	Must
FR-12	The system may provide an API for generating predictions for new product/store combinations.	Optional

Proposed KPIs

- Total Revenue: Sum of sales revenue over the selected period.
- Total Units Sold: Sum of quantities sold.
- Average Order Value (AOV): Total revenue divided by the number of valid orders, if order identifiers are available.

- Top and Bottom Products: Products ranked by a defined measure, such as revenue or units sold.
- Sales Growth: Percentage change in sales between comparable periods, when sufficient historical data exists.
- Prediction Error: Difference between predicted and actual values, evaluated using suitable regression metrics.

The exact KPI definitions should be finalized during implementation, depending on the available dataset fields.

3.4 Non-Functional Requirements

Non-functional requirements define how well the system should operate, including performance, security, usability, and reliability. The thresholds below are proposed acceptance targets, not existing tested results.

ID	Category	Requirement	Proposed Acceptance Criteria
NFR-01	Performance	Dashboard queries and filter updates should return results efficiently.	Common dashboard operations complete within 3 seconds on the agreed test dataset and environment.
NFR-02	Usability	The dashboard should be understandable and easy to navigate.	Users can locate KPIs and apply filters without technical assistance after a short introduction.

NFR-03	Security	Database access and sensitive configuration information should be protected.	Credentials are not hard-coded or committed to GitHub, and access is restricted where supported.
NFR-04	Data Quality	The system should detect relevant data-quality problems.	Missing values, duplicates, invalid dates, and inconsistent data types are checked and documented.
NFR-05	Reliability	The analytical workflow should handle common errors without silently producing misleading results.	Invalid inputs and processing errors produce clear messages or documented error handling.
NFR-06	Maintainability	The code should be organized and documented.	Code is separated into understandable components, with meaningful names and setup instructions.
NFR-07	Reproducibility	The analysis and model evaluation should be repeatable.	Dependencies, preprocessing steps, model settings, and evaluation

			procedures are documented.
NFR-08	Compatibility	The deliverables should work in the agreed development environment.	Database scripts, notebook, dashboard, and model can be run using documented software and configuration.
NFR-09	Scalability	The system design should support growth in the volume of sales records.	Query and processing performance is evaluated against the selected dataset size, with limitations documented.

3.5 Requirements Summary and Acceptance Criteria

The project will be considered functionally complete when the following conditions are met:

- ☐ Historical sales data is stored in a relational database and can be queried using the required dimensions.
- ☐ Sales KPIs and EDA results are calculated and documented.
- ☐ High- and low-performing products and temporal patterns are analyzed.
- ☐ A numerical prediction target is defined, and multiple regression approaches are compared.

- ☐ Model evaluation results are reported using appropriate metrics.
- ☐ An interactive dashboard presents the main findings.
- ☐ A business report explains the results and provides evidence-based recommendations.
- ☐ The SQL database scripts, analytical notebook, model, evaluation results, report, and project documentation are included in GitHub.
- ☐ The optional prediction API is implemented only if time and project requirements permit.

3.6 Assumptions and Constraints

- Historical sales data will be available in a usable digital format.
- The dataset will contain sufficient information to analyze sales across the requested dimensions; any missing dimensions will be documented.
- The numerical prediction target will be selected after examining the dataset. Possible targets include daily store sales or product-level sales, depending on the available fields.
- The regression model will be evaluated on data not used to train it. If the task is forecasting future periods, a time-based split should be considered to avoid data leakage.
- Seasonal analysis will depend on the length and granularity of the historical data.
- Business recommendations will be based on observed analytical results, not assumed outcomes.
- The prediction API is an optional extension, not a mandatory deliverable.